

Protozoa Lab Revision

* المراجعة دي
مختصرة + عالية العائد
للامتحان.

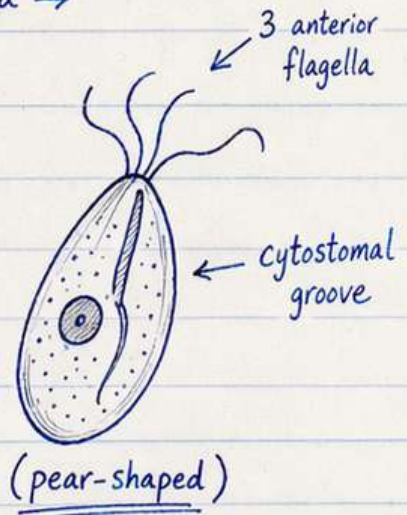
Lab 1 — Flagellates

1) Chilomastix sp.

- Classification: Kingdom Protozoa → Subkingdom Bilciliata →
Infrakingdom Excavata → Phylum Metamonada →
Class Retortamonada.

- Habitat: Intestine of man.
- Mode of living: commensal (nonpathogenic).
- Source: stool.
- Host: man.
- Lab features: 3 anterior flagella + 1 cytostomal groove.
- Mode of transmission: in eggs of nematodes.

مهم: nonpathogenic commensal.



2) Giardia sp.

- Classification: Kingdom Protozoa → Subkingdom Bilciliata →
Infrakingdom Excavata → Phylum Metamonada →
Class Diplomonada.
- Habitat: Small intestine of man.
- Mode of living: Parasite.
- Source: intestinal fluid (tissue).
- Host: man.
- Lab features: 8 flagella + 2 nuclei.
- Mode of transmission: fecal-oral route.
- Disease: Giardiasis → watery diarrhea.

Giardiasis = watery diarrhea.



2 nuclei
شكلهم زي
عينين

Quick compare

- Chilomastix: commensal, nonpathogenic.
- Giardia: parasite, causes diarrhea.
- Both belong to Excavata / Metamonada.

* للمراجعة
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للامتحان .

4 free
-flagella

1 posterior
flagellum

ata →
Class Kinetoplastea.

etoplastea.

شكلها شبه
الدبابيس أو
الإبر

Lab 2 — Apicomplexa

F.host = final host /
I.host = intermediate host

1) Plasmodium sp.

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Apicomplexa → Class Aconoidasida.
- **Habitat**: Intracellular blood of man.
- **Mode of living**: Parasite.
- **Source**: blood.
- **F.host**: female Anopheles sp. / **I.host**: man.
- **Lab features**: RBCs with dark area inside.
- **Mode of transmission**: By biting.
- **Disease**: Malaria.

Malaria.

2) Monocystis Lumbrici

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Apicomplexa → Class Conoidasida → Subclass Gregarinasina.
- **Habitat**: Seminal vesicle of earthworms.
- **Mode of living**: Parasite.
- **Disease**: Sterility.
- **SYZYGY phenomenon**: formation & differentiation of gametes in the same host cell.
- Gametes are Isogamy.

3) Hepatozoon sp.

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Apicomplexa → Class Conoidasida → Subclass Coccidiasina.
- **Habitat**: Seminal vesicle of earthworms.
- **Mode of living**: Parasite.
- **F.host**: Culex sp. / **I.host**: snake.
- **Disease**: Skeletal pain fever.
- It has **SYZYGY** phenomenon.
- It makes centripetal invagination.
- Gametes are Anisogamy.

Isogamy vs Anisogamy

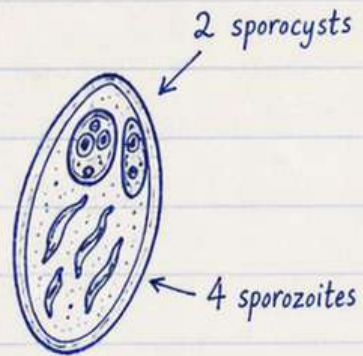
- Monocystis → Isogamy.
- Hepatozoon → Anisogamy.
- Both show **SYZYGY**.

Lab 2 — Apicomplexa (cont.)

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4) Sarcocystis sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Apicomplexa → Class Conoidasida → Subclass Coccidiasina.
- **Habitat:** Intestine of man.
- **Mode of living:** Parasite.
- **F.host:** Man / **I.host:** cattle or sheep.
- **Disease:** Disintegration of muscles, secretes toxic substance fatal to sheep and cattle.
- **Mode of transmission:** Eating raw or uncooked meat of sheep.
- $2 \times 4 = 2$ sporocysts & 4 sporozoites.



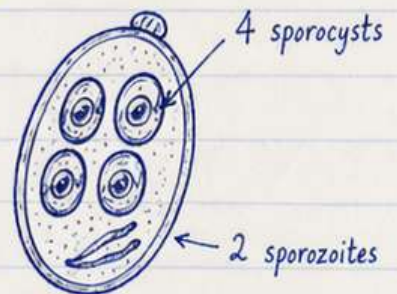
5) Toxoplasma sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Apicomplexa → Class Conoidasida → Subclass Coccidiasina.
- **Habitat:** Intestine of cat.
- **Mode of living:** Parasite.
- **F.host:** Cat / **I.host:** human.
- **Disease:** Toxoplasmosis → abortion in pregnant female cases.
- **Monoxenous life cycle.**
- **Cyst may be found in brain tissues.**



6) Eimeria sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Apicomplexa → Class Conoidasida → Subclass Coccidiasina.
- **Habitat:** Intestine of chicken and pigeons.
- **Mode of living:** Parasite.
- **F.host:** Chickens.
- **Disease:** Eimeriosis → weight loss, weakness, death.
- **Obligatory monoxenous life cycle.**
- $4 \times 2 = 4$ sporocysts & 2 sporozoites.



⇒ Exam trap ⇒

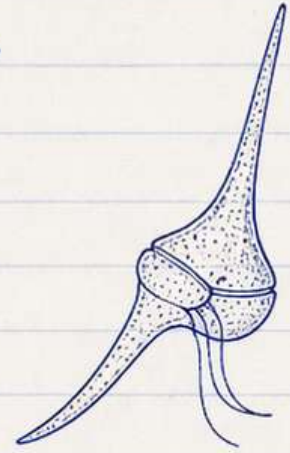
- Sarcocystis = 2×4 .
- Eimeria = 4×2 .
- Toxoplasma. مهم للحامل ☆

Lab 3 — Dinoflagellata & Ciliophora

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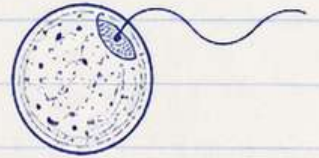
1) Ceratium sp.

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Dinoflagellata → Infraphylum Dinokaryota → Class Dinophyceae.
- **Habitat**: Marine and freshwater.
- **Lab feature**: single apical horn + two arms, with transverse flagellum and longitudinal flagellum.
- Colour of water becomes brown when present in high concentration.



2) Noctiluca sp.

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Dinoflagellata → Class Noctiluciphyceae.
- **Habitat**: Marine water.
- **Lab feature**: single flagellum + long food-capturing tentacle.
- **Bioluminescent**: when disturbed at night it gives soft green light due to luciferin granules in the cytoplasm.



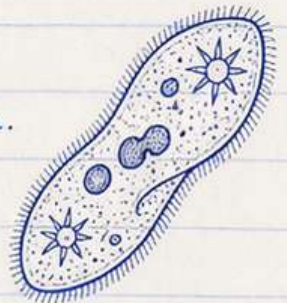
3) Blepharisma sp.

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Heterotrichea.
- **Habitat**: Marine water.
- **Mode of living**: free living.
- **Lab feature**: elongated thread-shaped macronucleus, pointed anterior end and rounded posterior end.



4) Paramecium sp.

- **Classification**: Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Oligohymenophorea.
- **Habitat**: freshwater oxygenated aquatic environments, feeding on bacteria.
- **Mode of living**: free living.
- **Lab feature**: slipper shape + 2 nuclei (macronucleus and micronucleus).
- Shows conjugation phenomenon.



Quick memory

- Noctiluca = night light.
- Paramecium = slipper shape.
- Ceratium = horned form.

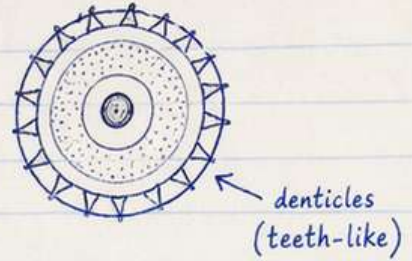


Lab 3 — Ciliophora (cont.)

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5) Trichodina sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Oligohymenophorea.
- **Habitat:** Ectoparasite on gills and skin of fishes.
- **Mode of living:** Ectoparasite.
- **Lab feature:** spherical body equipped with teeth-like **denticles** used for attachment.
- Heavy infection on gills may cause ulceration in host tissue.



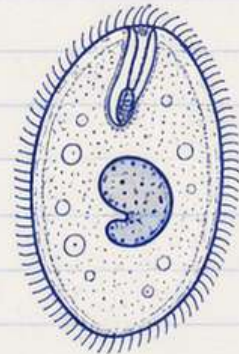
6) Ephelota sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Phyllopharyngea.
- **Habitat:** lives in ponds and attached to plants (sedentary).
- **Mode of living:** free living.
- **Lab feature:** bell-like body with a stalk at the base + branched macronucleus.



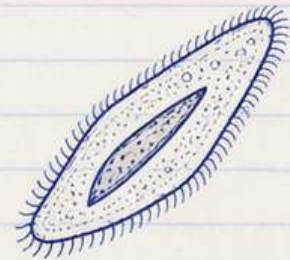
7) Balantidium coli

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Litostomatea.
- **Habitat:** Endoparasite in intestine of man.
- **Mode of living:** Endoparasite.
- **Lab feature:** **largest protozoan of man**, anterior cytostome, 2 lobed macronucleus.
- Trophozoite = motile vegetative stage in intestine.
- Cyst = infective stage, leaves with faeces and can be ingested by another host.



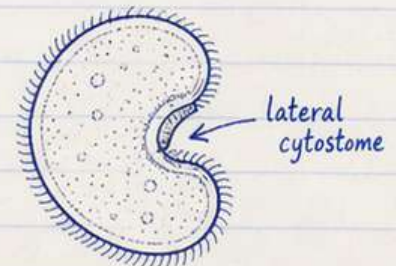
8) Epidinium sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Litostomatea.
- **Habitat:** fresh water.
- **Mode of living:** free living.
- **Lab feature:** elongated triangular body + triangular macronucleus with pointed and broad ends.



9) Nyctotherus sp.

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata → Infrakingdom Alveolata → Phylum Ciliophora → Class Armophorea.
- **Habitat:** colon of toads and frogs.
- **Mode of living:** Endoparasite.
- **Lab feature:** kidney-shaped body with lateral cytostome.
- Tadpoles are infected by eating faeces containing the cyst (infective stage).



Infective stages

- Balantidium : cyst.
- Nyctotherus : cyst.

تم إنشاء هذا البورصة المتهري عبد المنعم وفتح 2023
وبالاستعانة بأولاد الزكاء الاصطفاي 2026

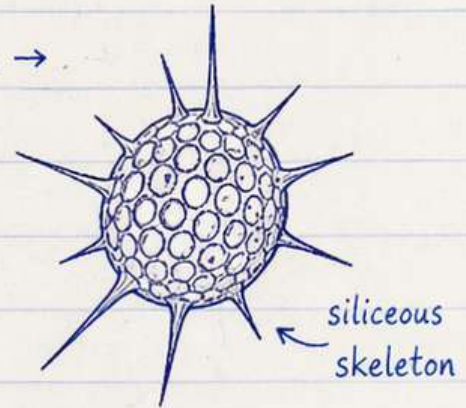
Protozoa Lab Revision

Lab 4 — Rhizaria & Amoebozoa

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للإمتحان.

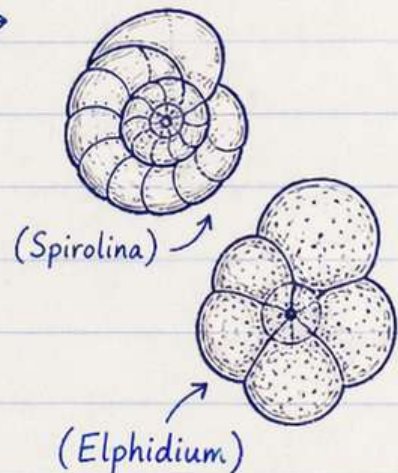
1) Radiolarians

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata →
Infrakingdom Rhizaria → Phylum Radiozoa →
Class Polycystinea.
- Produce **siliceous skeletons**.
- Include the majority of fossils.
- Marine holoplanktonic protozoa widely distributed in the oceans.



2) Foraminiferans / Elphidium / Polystomella / Spirolina

- **Classification:** Kingdom Protozoa → Subkingdom Bilciliata →
Infrakingdom Rhizaria → Phylum Foraminifera →
Class Foraminiferidea.
- Foraminifera = "hole bearers".
- Large phylum of marine amoeboid protists.
- Typically produce a test or shell.
- Shells are made of calcium carbonate (CaCO_3) or agglutinated sediment particles.



Radiolarians = siliceous / Foraminifera = calcareous test

3) Entamoeba histolytica & Entamoeba coli

- **Classification:** Kingdom Protozoa → Subkingdom Sarcomastigota →
Phylum Amoebozoa → Class Archamoeba.
- Important point in this sheet : focus on classification.

المهم
هنا
التصنيف

Quick compare

- Radiolarians → silica.
- Foraminiferans → CaCO_3 shell / test.
- Entamoeba here → Amoebozoa.

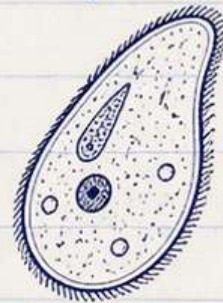


Lab 4 — Final review

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للامتحان .

4) Opalina sp. & Protooplinia sp.

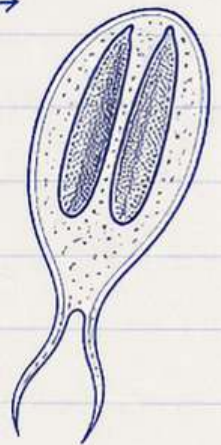
- **Classification:** Kingdom Protozoa → Subkingdom Sarcomastigota → Phylum Opalozoa → Class Opalineae.
- **Mode of living:** Commensal.
- **Habitat:** intestines of frogs and toads.



← frog / toad intestine

5) Henneguya suprabranchiae

- **Classification:** Kingdom Protozoa → Subkingdom Sarcomastigota → Phylum Myxozoa → Class Myxosporea.
- **Mode of living:** Endoparasite.
- **Habitat:** infects the hyaline cartilage of the suprabranchial organs of catfish.
- **Lab feature:** two identical elongate polar capsules.
- Posteriorly there are two thin caudal processes.



Rapid exam review

- Giardia → watery diarrhea.
- Trichomonas → sexual transmission.
- Trypanosoma → Tse-tse fly → sleeping sickness.
- Leishmania → sand fly.
- Plasmodium → female Anopheles → malaria.
- Toxoplasma → cat / pregnancy risk.
- Balantidium coli → largest protozoan of man.
- Radiolarians → siliceous skeleton.
- Foraminiferans → CaCO_3 test.

host + transmission + disease + lab feature

Done — بالتوفيق فى الامتحان —

خاتمة وتوقيع

تحية إنشاء هذا بوريسطه المهدى عبد الله من رفعة 2023 وبالبستعانة بأدوات الزكاء الاصطناعي 2026

إذا رغبت في إعداد ملخصات أكاديمية احترافية بهذا المستوى، أو تعلم كيفية
تصميم ملفات أكثر جمالاً وتنظيماً واحترافية لأي مادة، فتواصل معي وسأساعدك
على الوصول إلى نتيجة أرقى وأقوى بكثير.

للتواصل: zootopiaclub.studio/contact